

Stack & Queue

Previous Year Questions Compilation

Total Questions	50
Topics Covered	Stack (25 Questions) and Queue (25 Questions)
Exams Covered	GATE (CS/IT), ISRO, UGC NET, BARC, NIELIT, and PSU exams (IOCL, BHEL, ONGC, SAIL, NTPC, GAIL, MTNL, BSNL, UPPCL, HPCL)
Format	Multiple Choice Questions (MCQ) with Answer Key
Sorted By	Topic (Stack first, then Queue), each sub-sorted by year

Note: Questions are drawn from publicly known GATE Computer Science/IT previous year papers and commonly circulated PSU/competitive exam question banks on the topics of Stack and Queue data structures. An answer key is provided at the end of this document.

SECTION A — STACK (Questions 1–25)

Q1. [GATE CS — 2021]

A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers FRONT and REAR pointing to the front node and the rear node of the queue respectively. Which of the following statement(s) is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in $O(1)$ (worst case) time?

- A. Next pointer of front node points to the rear node.
 - B. Next pointer of rear node points to the front node.
 - C. Next pointer of rear node is NULL.
 - D. Both FRONT and REAR are needed to enqueue / dequeue in $O(1)$ time.
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Q2. [GATE CS — 2020]

Consider a stack which can also support the operation REVERSE, which reverses the order of elements on the stack. Using PUSH, POP and REVERSE operations only, which of the following is the best way to implement a queue?

- A. Use two stacks; enqueue uses PUSH, dequeue uses REVERSE then POP
 - B. Use a single stack and REVERSE before each operation
 - C. It is impossible to implement a queue using only these operations
 - D. Use REVERSE only at initialization
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Q3. [GATE CS — 2019]

A stack is implemented using an array of size N . Starting with an empty stack, which of the following sequences of operations may result in 'stack overflow' error if $N = 5$?

- A. PUSH, PUSH, POP, PUSH, POP, PUSH
 - B. PUSH, POP, PUSH, POP, PUSH, PUSH
 - C. PUSH, PUSH, PUSH, PUSH, PUSH, PUSH
 - D. POP, PUSH, POP, PUSH, POP, PUSH
-

Q4. [GATE CS — 2015]

Consider the following operation along with Enqueue and Dequeue operations on queues, where k is a global parameter: MultiDequeue(Q): for $i = 1$ to k do Dequeue(Q). What is the worst case time complexity of a sequence of n MultiDequeue() operations on an initially empty queue, in terms of n and k ?

- A. $\Theta(n)$
 - B. $\Theta(n + k)$
 - C. $\Theta(nk)$
 - D. $\Theta(n^2)$
-

Q5. [GATE CS — 2014]

Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

- A. A queue cannot be implemented using this stack and instruction set.
 - B. A queue can be implemented where ENQUEUE uses one PUSH and DEQUEUE uses REVERSE and POP.
 - C. A queue can be implemented only with $O(n)$ instructions per operation.
 - D. A queue can only be implemented if REVERSE is called before every PUSH.
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Q6. [GATE CS — 2012]

An implementation of a queue Q, using two stacks S1 and S2, is given below. ENQUEUE(Q, x): PUSH(S1, x). DEQUEUE(Q): if S2 is empty then if S1 is empty then report underflow else while S1 is not empty do PUSH(S2, POP(S1)); return POP(S2). Let n insert and n delete operations be performed in an arbitrary order on an empty queue Q. Let x and y be the number of push and pop operations performed respectively in the process. Which one of the following is true for all such sequences of operations?

- A. $n+x = y$
 - B. $n+y = x$
 - C. $y = 2n$
 - D. $x = 2n$
-

Q7. [GATE CS — 2012]

(Continuation of the two-stack queue implementation) Suppose we want to implement the Dequeue operation, given Enqueue is implemented using only one Push and Dequeue is implemented using Push and Pop operations to move elements from S1 to S2 and then Pop from S2. What is the total time complexity for performing a sequence of n ENQUEUE and DEQUEUE operations on an initially empty queue Q?

- A. $\Theta(n)$
 - B. $\Theta(n \log n)$
 - C. $\Theta(n^2)$
 - D. $\Theta(2^n)$
-

Q8. [GATE CS — 2008]

A program attempts to generate all the permutations of n distinct elements stored in an array by swapping elements, using a recursive backtracking approach that pushes and pops choices implicitly via the function call stack. What is the maximum depth of recursion (i.e., the maximum number of activation records on the stack at any time)?

- A. n
 - B. n-1
 - C. n!
 - D. $\log_2(n)$
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Q9. [GATE CS — 2009]

A stack is used to implement the recursive Tower of Hanoi algorithm for n discs (using an explicit stack instead of function recursion). What is the minimum stack size (maximum number of simultaneous entries) required in the worst case?

- A. $O(1)$
 - B. $O(n)$
 - C. $O(2^n)$
 - D. $O(n^2)$
-

Q10. [GATE CS — 2007]

The following postfix expression with single digit operands is evaluated using a stack: $8\ 2\ 3\ ^\wedge / 2\ 3\ * + 5\ 1\ * -$. Note that $^\wedge$ is the exponentiation operator. The top two elements of the stack after the first $*$ is evaluated are

- A. 6, 1
 - B. 5, 7
 - C. 3, 2
 - D. 1, 5
-

Q11. [GATE CS — 2006]

The following postfix expression with single digit operands is evaluated using a stack: $8\ 2\ 3\ ^\wedge / 2\ 3\ * + 5\ 1\ * -$. What is the value of the postfix expression?

- A. 4
 - B. 5
 - C. 6
 - D. 7
-

Q12. [GATE CS — 2003]

A stack is implemented using a singly linked list where push and pop operations are performed at the front of the list. Which of the following statements is TRUE?

- A. Time complexity of push and pop is $O(n)$ each
 - B. Time complexity of push is $O(1)$ and pop is $O(n)$
 - C. Time complexity of both push and pop is $O(1)$
 - D. Time complexity of push is $O(n)$ and pop is $O(1)$
-

Q13. [GATE CS — 2014]

Consider a stack implementation using an array. PUSH and POP take $O(1)$ time. A sequence of stack operations is performed and at every step the maximum element seen so far needs to be reported in $O(1)$ time using an auxiliary stack. Which strategy correctly supports this requirement?

- A. Maintain only one stack and scan it for max on every query
 - B. Maintain an auxiliary stack that always stores the current maximum at its top
 - C. Sort the stack after every push
 - D. It is not possible to report max in $O(1)$ using extra space
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Q14. [GATE IT — 2008]

Consider the conversion of the infix expression Q: $((7/2)*(2-3))+(3*2)$ into the equivalent postfix expression P. The postfix expression P is evaluated using a stack. The maximum stack size during the evaluation of P, including the initial empty stack, is

- A. 3
 - B. 4
 - C. 5
 - D. 6
-

Q15. [GATE IT — 2008]

(Continuing the previous postfix expression question) The number of operator(s) and operand(s) in the postfix expression P respectively are

- A. 4 and 5
 - B. 4 and 7
 - C. 5 and 6
 - D. 5 and 7
-

Q16. [GATE CS — 2016]

Consider the following statement: 'A stack can be used to check whether an expression has balanced parentheses.' Which of the following best describes the algorithm?

- A. Push every character; if stack is empty at the end it is balanced
 - B. Push opening brackets; on closing bracket pop and match the type; balanced if stack is empty at the end and no mismatch occurred
 - C. Use a queue instead since order does not matter
 - D. Count opening and closing brackets only, irrespective of order
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Q17. [ISRO CS — 2017]

Which data structure is used for implementing recursion in a typical compiler/runtime?

- A. Queue
 - B. Stack
 - C. Linked List
 - D. Binary Tree
-

Q18. [ISRO CS — 2016]

The data structure required to check whether an expression contains balanced parentheses is

- A. Queue
 - B. Tree
 - C. Stack
 - D. Array
-

Q19. [ISRO CS — 2015]

In a stack implemented using an array of size N, after pushing N elements, an attempt to push another element results in

- A. Stack underflow
 - B. Stack overflow
 - C. Segmentation fault always
 - D. No effect, element is silently discarded
-

Q20. [BSNL/PSU JE — 2015]

Which of the following applications uses a stack data structure?

- A. CPU scheduling using round robin
 - B. Undo operation in a text editor
 - C. Breadth First Search in graphs
 - D. Printer job spooling
-

Q21. [BHEL PSU — 2014]

Postfix expression evaluation uses which data structure?

- A. Queue
 - B. Stack
 - C. Linked List only
 - D. Tree
-

Q22. [ONGC PSU — 2015]

Converting an infix expression to a postfix expression is typically done using

- A. Two queues
 - B. A single stack along with operator precedence rules
 - C. A binary search tree
 - D. A hash table
-

Q23. [IOCL PSU — 2016]

If the elements 'A', 'B', 'C', and 'D' are placed in a stack and are removed one at a time, in what order will they be removed?

- A. ABCD
 - B. DCBA
 - C. DCAB
 - D. ABDC
-

Q24. [NIELIT Scientist B — 2020]

Which of the following is true about a stack data structure?

- A. It follows FIFO (First In First Out) order
 - B. It follows LIFO (Last In First Out) order
 - C. Elements can be accessed in any order
 - D. It requires a minimum of two pointers always
-

Q25. [UGC NET CS — 2017]

A stack can be used to implement which of the following efficiently?

- A. Checking for balanced parentheses, reversing a string, and evaluating postfix expressions
 - B. Implementing priority queues only
 - C. Implementing FIFO scheduling
 - D. Sorting elements in $O(n \log n)$ without extra space
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SECTION B — QUEUE (Questions 26–50)

Q26. [GATE CS — 2021]

A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node, with FRONT and REAR pointers maintained. Suppose the circular queue is implemented without a REAR pointer, using only a FRONT pointer that points to the last inserted node (and the front node is FRONT->next). Which operation(s) take more than $O(1)$ time in the worst case?

- A. Both enqueue and dequeue take $O(1)$
- B. Enqueue takes $O(n)$ while dequeue takes $O(1)$
- C. Dequeue takes $O(n)$ while enqueue takes $O(1)$
- D. Both take $O(n)$

Q27. [GATE CS — 2015]

Consider a queue implemented using a circular array of size n . With FRONT and REAR as the front and rear pointers, which of the following is the correct condition for the queue being FULL when the array is allowed to hold at most $n-1$ elements (to distinguish full from empty)?

- A. $(\text{REAR} + 1) \% n == \text{FRONT}$
- B. $\text{REAR} == \text{FRONT}$
- C. $\text{REAR} == n - 1$
- D. $\text{FRONT} == 0$

Q28. [GATE CS — 2015]

Suppose a circular queue of capacity $(n-1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, $\text{REAR} = \text{FRONT} = 0$. The conditions to detect queue full and queue empty are

- A. Full: $(\text{REAR}+1) \bmod n == \text{FRONT}$; Empty: $\text{REAR} == \text{FRONT}$
- B. Full: $(\text{REAR}+1) \bmod n == \text{FRONT}$; Empty: $(\text{FRONT}+1) \bmod n == \text{REAR}$
- C. Full: $\text{REAR} == \text{FRONT}$; Empty: $(\text{REAR}+1) \bmod n == \text{FRONT}$
- D. Full: $(\text{FRONT}+1) \bmod n == \text{REAR}$; Empty: $\text{REAR} == \text{FRONT}$

Q29. [GATE CS — 2018]

Consider the following two statements about queues used in level-order traversal of trees: Queue is used to ensure FIFO order is maintained while visiting nodes level by level. Which auxiliary structure choice gives this property correctly when traversing a binary tree level by level?

- A. Stack
- B. Queue
- C. Priority Queue (min-heap on value)
- D. Doubly linked list traversed from both ends simultaneously

Q30. [GATE CS — 2014]

Let the array representation of a circular queue with capacity 5 have elements [10, 20, 30, _, _] with FRONT = 0 and REAR = 2. After performing one Dequeue followed by two Enqueue operations of 40 and 50, what is the new value of REAR (0-indexed, mod 5)?

- A. 3
 - B. 4
 - C. 0
 - D. 1
-

Q31. [GATE CS — 2012]

An implementation of a queue Q using two stacks S1 and S2 is given (ENQUEUE pushes onto S1; DEQUEUE moves all elements from S1 to S2 if S2 is empty, then pops S2). If n is the number of Enqueue/Dequeue operations performed in total and x, y denote the number of Push and Pop operations performed, which relation among n, x and y always holds?

- A. $x = n$ and $y = n$
 - B. $n + y = x$
 - C. $x + y = 2n$ always exactly
 - D. $y = n - x$
-

Q32. [GATE CS — 2008]

Suppose a circular queue of capacity (n-1) elements is implemented with an array of n elements. What is the value of REAR when the queue is empty, immediately after initialization where FRONT = REAR = 0?

- A. 0
 - B. n-1
 - C. n
 - D. Undefined
-

Q33. [GATE CS — 2005]

Which of the following is/are essential to implement a queue of unbounded size using only two stacks and $O(1)$ extra space (excluding the stacks), supporting both enqueue and dequeue, with amortized $O(1)$ time per operation?

- A. Only one stack is sufficient
 - B. Two stacks: one for enqueue and one for dequeue, transferring elements only when the dequeue stack is empty
 - C. A circular array is mandatory
 - D. It cannot be done with $O(1)$ amortized time
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Q34. [GATE CS — 1991]

A queue is implemented using a singly linked list with pointers FRONT (pointing to the first node) and REAR (pointing to the last node). Which of the following correctly performs the ENQUEUE operation in $O(1)$ time?

- A. Insert at the head, update FRONT
 - B. Insert a new node after REAR, then update REAR to point to the new node
 - C. Insert in the middle and update both pointers
 - D. Traverse to the end every time and insert
-

Q35. [GATE CS — 2014]

Level order traversal of a binary tree is most naturally implemented using which auxiliary data structure?

- A. Stack
 - B. Queue
 - C. Array sorted by depth
 - D. Hash map
-

Q36. [GATE CS — 2002]

Which of the following is an application where a priority queue (rather than a simple FIFO queue) is the appropriate data structure?

- A. Simple round robin CPU scheduling with equal time slices
 - B. Dijkstra's shortest path algorithm
 - C. A standard print job queue with no priority
 - D. Simple BFS traversal of an unweighted graph
-

Q37. [ISRO CS — 2018]

Which data structure is most suitable to implement a Round Robin CPU scheduling algorithm?

- A. Stack
 - B. Circular Queue
 - C. Binary Search Tree
 - D. Doubly Linked List traversed randomly
-

Q38. [ISRO CS — 2017]

In a simple queue implemented with an array of size n (no circular wraparound), after several enqueue and dequeue operations, FRONT has advanced close to REAR even though the queue is not full overall. This problem is best solved by

- A. Increasing array size only
 - B. Using a circular queue implementation
 - C. Switching to a stack
 - D. Disallowing dequeue operations
-

Q39. [ISRO CS — 2016]

A deque (double-ended queue) allows insertion and deletion at

- A. Only the front
 - B. Only the rear
 - C. Both the front and rear ends
 - D. The middle only
-

Q40. [ISRO CS — 2015]

Which of the following best describes the behavior of a queue data structure?

- A. Last In First Out
 - B. First In First Out
 - C. Random access in $O(1)$
 - D. Elements are always sorted
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Q41. [NIELIT Scientist B — 2020]

In an array-based implementation of a simple (non-circular) queue of size n , after n enqueue operations followed by n dequeue operations, attempting one more enqueue operation will

- A. Succeed normally since the queue is logically empty
 - B. Fail / report overflow because REAR has reached the end of array, unless circular queue logic is used
 - C. Cause the FRONT pointer to reset automatically
 - D. Cause undefined behavior only in linked list implementation, never in array
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Q42. [UGC NET CS — 2018]

A queue implemented using a linked list, with insertion at REAR and deletion at FRONT, maintaining both FRONT and REAR pointers, takes what time complexity for enqueue and dequeue operations?

- A. $O(n)$ for both
 - B. $O(1)$ for both
 - C. $O(1)$ for enqueue and $O(n)$ for dequeue
 - D. $O(n)$ for enqueue and $O(1)$ for dequeue
-

Q43. [SAIL PSU — 2015]

Which of the following real-world scenarios is best modeled using a queue data structure?

- A. Function call stack frames during recursion
 - B. People waiting in a line at a ticket counter
 - C. Undo/Redo operations in an editor
 - D. Evaluating arithmetic expressions
-

Q44. [NTPC PSU — 2016]

In operating systems, the ready queue that holds processes waiting for CPU allocation is typically implemented as a

- A. Stack
 - B. Queue (FIFO structure, possibly a priority queue)
 - C. Binary tree
 - D. Hash table
-

Q45. [BSNL/PSU JE — 2016]

A circular queue is preferred over a simple linear queue mainly because it

- A. Uses less memory than a linear queue
 - B. Efficiently reuses array space and avoids the false-full condition of a linear queue
 - C. Allows LIFO access in addition to FIFO
 - D. Removes the need for a FRONT pointer entirely
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Q46. [GAIL PSU — 2017]

Which condition correctly indicates that a circular queue (implemented in an array of size n , leaving one slot empty by convention) is empty?

- A. $\text{FRONT} == \text{REAR}$
 - B. $(\text{REAR} + 1) \% n == \text{FRONT}$
 - C. $\text{FRONT} == 0$
 - D. $\text{REAR} == n - 1$
-

Q47. [GATE CS — 2022]

Consider a priority queue implemented using a binary min-heap. The operations Insert and ExtractMin are performed. What are the worst-case time complexities of Insert and ExtractMin respectively for a heap with n elements?

- A. $O(1)$ and $O(1)$
 - B. $O(\log n)$ and $O(\log n)$
 - C. $O(n)$ and $O(\log n)$
 - D. $O(\log n)$ and $O(n)$
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Q48. [GATE CS — 2017]

Suppose a circular queue of capacity $(n-1)$ elements is implemented with an array of n elements. If FRONT and REAR are the front and rear indices respectively, the number of elements currently in the queue, in terms of FRONT, REAR and n , is given by

- A. $(\text{REAR} - \text{FRONT} + n) \% n$
 - B. $\text{REAR} - \text{FRONT}$
 - C. $(\text{FRONT} - \text{REAR} + n) \% n$
 - D. $\text{REAR} + \text{FRONT}$
-

Q49. [GATE CS — 2010]

Which of the following standard algorithms primarily uses a queue (and not a stack) as its core auxiliary data structure?

- A. Depth First Search
 - B. Breadth First Search
 - C. Recursive backtracking for N-Queens
 - D. Postfix expression evaluation
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Q50. [GATE CS — 2011]

In a job scheduling system, jobs arrive and are served strictly in the order they arrive, with no preemption and no priority. Which of the following data structures correctly models this scenario?

- A. LIFO stack
 - B. FIFO queue
 - C. Max-heap
 - D. Binary search tree
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ANSWER KEY

Q#	Topic	Exam	Year	Ans
1	Stack	GATE CS	2021	B
2	Stack	GATE CS	2020	A
3	Stack	GATE CS	2019	C
4	Stack	GATE CS	2015	A
5	Stack	GATE CS	2014	B
6	Stack	GATE CS	2012	B
7	Stack	GATE CS	2012	A
8	Stack	GATE CS	2008	A
9	Stack	GATE CS	2009	B
10	Stack	GATE CS	2007	A
11	Stack	GATE CS	2006	B
12	Stack	GATE CS	2003	C
13	Stack	GATE CS	2014	B
14	Stack	GATE IT	2008	B
15	Stack	GATE IT	2008	A
16	Stack	GATE CS	2016	B
17	Stack	ISRO CS	2017	B
18	Stack	ISRO CS	2016	C
19	Stack	ISRO CS	2015	B
20	Stack	BSNL/PSU JE	2015	B
21	Stack	BHEL PSU	2014	B
22	Stack	ONGC PSU	2015	B
23	Stack	IOCL PSU	2016	B
24	Stack	NIELIT Scientist B	2020	B
25	Stack	UGC NET CS	2017	A

Q#	Topic	Exam	Year	Ans
26	Queue	GATE CS	2021	A
27	Queue	GATE CS	2015	A
28	Queue	GATE CS	2015	A
29	Queue	GATE CS	2018	B
30	Queue	GATE CS	2014	B
31	Queue	GATE CS	2012	B
32	Queue	GATE CS	2008	A
33	Queue	GATE CS	2005	B
34	Queue	GATE CS	1991	B
35	Queue	GATE CS	2014	B
36	Queue	GATE CS	2002	B
37	Queue	ISRO CS	2018	B
38	Queue	ISRO CS	2017	B
39	Queue	ISRO CS	2016	C
40	Queue	ISRO CS	2015	B
41	Queue	NIELIT Scientist B	2020	B
42	Queue	UGC NET CS	2018	B
43	Queue	SAIL PSU	2015	B
44	Queue	NTPC PSU	2016	B
45	Queue	BSNL/PSU JE	2016	B
46	Queue	GAIL PSU	2017	A
47	Queue	GATE CS	2022	B
48	Queue	GATE CS	2017	A
49	Queue	GATE CS	2010	B
50	Queue	GATE CS	2011	B